

RESEARCH ARTICLE

Deep Recurrent Framework for Accurate Estimation of SOC and SOH in Lithium-Ion Battery

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ABSTRACT The rapid growth of electric vehicles and renewable energy storage has made accurate battery state estimation critical for safe and efficient operation. This work presents two hybrid deep learning frameworks for estimating the State of Health (SOH) and State of Charge (SOC) of lithium-ion batteries: an Incremental Capacity Analysis–Gated Recurrent Unit (ICAGRU) model for SOH, and an Extended Kalman Filter–Gated Recurrent Unit (EKFGRU) model for SOC. The ICAGRU architecture fuses electrochemical features extracted via incremental capacity analysis with a GRU networks to capture degradation signatures and temporal aging dynamics, enabling enhanced SOH estimation. The EKFGRU framework couples an extended Kalman filter with deep learning to achieve robust SOC tracking under dynamic load conditions by compensating for model uncertainties. Both frameworks are validated on NASA Ames battery datasets (B0005, B0033, B0036) and a dynamic load profile, covering diverse aging behaviors and realistic operating scenarios. The proposed ICAGRU model achieves an SOH estimation error of approximately 2%, while the EKFGRU maintains SOC estimation errors below 2%, representing substantial improvements over conventional methods and standard GRU-based approaches. By combining physics based modelling with data-driven learning, these hybrid architectures provide superior accuracy, generalization, and interpretability, advancing battery management systems for enhanced safety and performance and establishing a foundation for future intelligent energy storage solutions.

INDEX TERMS Battery management system, state of health, state of charge, incremental capacity analysis, extended Kalman filter, gated recurrent unit, deep learning, lithium-ion battery, hybrid modelling, electrochemical features.

I. INTRODUCTION

The global transition towards sustainable transportation and integration of renewable energy has made lithium-ion batteries the cornerstone technology for energy storage applications. With the electric vehicle market projected to reach 145 million units by 2030 and grid-scale energy storage deployments experiencing exponential growth, the demand for sophisticated battery management systems (BMS) has never been more critical [1]. At the heart of effective battery management lies the accurate estimation of two fundamental state variables: State of Health (SOH), which quantifies

the battery's aging condition relative to its initial capacity, and State of Charge (SOC), which indicates the remaining available energy.

The estimation of battery states presents formidable technical challenges that stem from the complex, non-linear electrochemical processes occurring within the cell. Unlike conventional systems where states can be directly measured, battery internal states must be inferred from available measurements such as terminal voltage, current, and surface temperature. This indirect estimation problem is compounded by the highly non-linear relationship between measurable quantities and internal states, the time-varying nature of battery parameters due to aging, the influence of operating conditions, including temperature and current

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rates, and the presence of measurement noise and modelling uncertainties [2].

Traditional estimation methods suffer from fundamental limitations that restrict their accuracy and applicability. Coulomb counting accumulates errors over time due to measurement inaccuracies and requires periodic recalibration. Voltage-based methods face challenges with the flat voltage plateaus characteristic of lithium-ion batteries. Impedance-based techniques require specialized excitation signals and measurement equipment that are not suitable for real-time implementation [3]. Model-based techniques, particularly the Extended Kalman Filter (EKF), have gained widespread adoption due to their ability to handle non-linear systems while providing uncertainty quantification [4]. However, these approaches rely heavily on accurate battery models, which are challenging to maintain throughout the battery's operational life.

Recent advances in machine learning have opened new avenues for battery state estimation [5]. Recurrent Neural Networks (RNNs) and their advanced variants, including Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), have demonstrated remarkable success in time-series estimation tasks [6]. However, purely data-driven approaches face their challenges, including limited interpretability and poor generalization when operating conditions deviate from training data distribution [7]. For instance, SVR trained on first-life voltage measurements has shown promise in predicting second-life battery state-of-health, demonstrating that accessible electrical signals carry sufficient degradation information for cross-stage generalization [8]. Standard recurrent models such as GRU and LSTM process voltage and current sequences but lack an internal physical model of battery dynamics. Under dynamic loads with rapid current reversals characteristic of electric vehicle drive cycles (e.g., regenerative braking), terminal voltage exhibits complex transient behavior driven by ohmic drops, charge transfer overpotential, and diffusion effects occurring at distinct time scales (1 10–100 s for charge transfer, 2 100–1000 s for diffusion). Purely data-driven models must implicitly learn these multi-timescale dynamics from training data alone, leading to a systematic estimation lag of 2 to 3% SOC during rapid current transitions and elevated errors at SOC extremes where the OCV-SOC relationship is highly non-linear. These limitations motivate the integration of physics-based filtering with deep learning, where the model-based component explicitly tracks known battery dynamics while the neural network learns to compensate for residual model uncertainties.

A. COMPARISON WITH EXISTING HYBRID SOC/SOH ESTIMATORS

The recent literature has explored various combinations of physics-based and data-driven methods for battery state estimation. For SOH estimation, ICA-based features have been combined with machine learning models, including

SVR, convolutional neural networks, and gradient boosting methods. For SOC estimation, EKF-neural network hybrids have been proposed, where neural networks serve to estimate EKF parameters, provide voltage corrections, or replace the measurement model entirely. Physics-informed neural networks (PINNs) have also been applied to battery degradation modelling, embedding governing equations directly into the loss function.

However, several gaps remain in the existing literature. First, no prior work combines the full set of ICA-derived electrochemical features (peak intensity, position, width, area, skewness, kurtosis) with recurrent neural networks augmented by attention mechanisms for temporal degradation modelling. Existing ICA-ML approaches typically use only peak height or position as features and employ static models (SVR, RF) that cannot capture degradation trajectory dynamics. Second, existing EKF-RNN hybrids typically employ a simple cascaded architecture where the neural network either pre-processes features for the EKF or post-processes its output. Our three-module architecture—comprising an adaptive EKF, a residual correction network, and a GRU refinement network with learned fusion weights—represents a deeper integration that enables uncertainty-aware information combination. Third, to our knowledge, no prior framework addresses both SOH and SOC estimation within a unified hybrid design philosophy, which we argue provides complementary insights for comprehensive battery management.

Recent advances in SOC estimation methods include hybrid observer-estimator approaches [9], knowledge-informed transfer learning frameworks [10], and pattern recognition-based techniques for battery state estimation [11], which further motivate the need for architectures that can synergize physics-based knowledge with adaptive learning capabilities.

The integration of physics-based knowledge with data-driven methods presents a compelling paradigm that addresses these limitations. Physics-informed machine learning has gained traction in various scientific domains, demonstrating that incorporating domain knowledge can significantly improve model performance and generalization [12], [13], [14], [15], [16], [17], [18], [19]. In battery state estimation, Incremental Capacity Analysis (ICA) has emerged as a powerful diagnostic tool, transforming the capacity-voltage relationship into differential form that amplifies degradation signatures [20]. These electrochemical signatures provide valuable features that can enhance machine learning models by encoding physical degradation mechanisms. Our SHAP analysis (Section V-D) confirms that the six features contribute complementary degradation information: while peak intensity dominates (mean SHAP = 0.324), removing any individual secondary feature (peak position, FWHM, or curve area) increases SOH estimation RMSE by 12–18%. Specifically, peak broadening (FWHM) captures growing electrode heterogeneity not visible in height measurements alone, while skewness and kurtosis detect

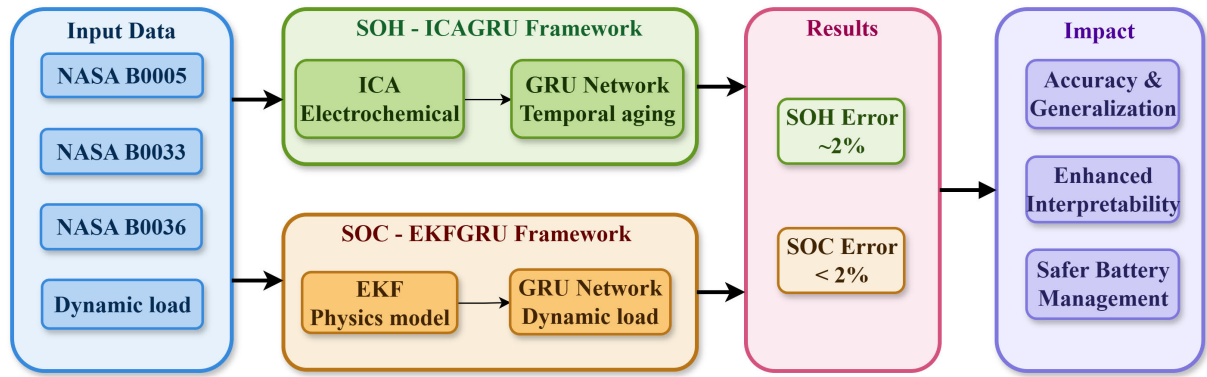


FIGURE 1. Proposed hybrid framework for SOH and SOC estimation of lithium-ion batteries.

asymmetric degradation patterns characteristic of lithium plating onset a critical safety concern that single-feature approaches cannot identify.

This paper addresses the critical need for accurate battery state estimation by proposing two novel hybrid frameworks that synergistically combine physics-based feature extraction with deep learning architectures. The ICAGRU model integrates Incremental Capacity Analysis with Gated Recurrent Unit networks for enhanced SOH estimation, capturing both electrochemical degradation mechanisms and complex temporal aging patterns. The EKFGRU framework combines an Extended Kalman Filter with GRU networks to improve SOC tracking, leveraging model-based estimation as a foundation while employing deep learning to compensate for model uncertainties under dynamic load conditions. As illustrated in Figure 1, both frameworks are validated on NASA Ames battery datasets (B0005, B0033, B0036, Dynamic load) and achieve estimation errors below 2%, demonstrating significant improvements over conventional methods and standard GRU-based approaches.

The main contributions of this work include:

- 1) Development of novel hybrid architectures that effectively integrate physics-based knowledge with deep learning;
- 2) Comprehensive feature engineering methodology capturing electrochemical signatures through ICA;
- 3) Extensive experimental validation using NASA battery datasets representing different degradation scenarios;
- 4) Detailed comparative analysis demonstrating advantages over conventional approaches
- 5) Practical implementation considerations for real-world deployment.

II. BATTERY MODELLING AND FEATURE EXTRACTION

A. LITHIUM-ION BATTERY FUNDAMENTALS

Understanding the electrochemical processes within lithium-ion batteries is crucial for developing effective state estimation algorithms [21]. During discharge, lithium ions migrate from the negative electrode (typically graphite)

through the electrolyte to the positive electrode (such as LiCoO_2 , LiMn_2O_4 , or LiFePO_4), while electrons flow through the external circuit [22]. This process reverses during charging, with the battery's performance being governed by complex interactions between thermodynamics, kinetics, and transport phenomena.

The battery's terminal voltage represents a superposition of multiple components:

$$V_{\text{terminal}} = OCV(SOC, T) - V_{\text{ohmic}} - V_{\text{ct}} - V_{\text{diff}} - V_{\text{solid}} \quad (1)$$

where $OCV(SOC, T)$ is the open-circuit voltage dependent on SOC and temperature T , V_{ohmic} represents ohmic losses in current collectors and electrolyte, V_{ct} captures charge transfer overpotentials at electrode-electrolyte interfaces, V_{diff} accounts for concentration gradients in electrolyte and solid phases, and V_{solid} represents solid-phase diffusion limitations within electrode particles.

Battery aging manifests through multiple degradation mechanisms that occur simultaneously and interact in complex ways [23]. Loss of lithium inventory (LLI) occurs due to side reactions that consume cyclable lithium, such as the growth of solid-electrolyte interphase (SEI) and lithium plating. Loss of active material (LAM) occurs through mechanical degradation including particle cracking, electrical isolation, and binder decomposition. Impedance growth is attributed to SEI thickening, current collector corrosion, and electrolyte decomposition. These mechanisms progress at different rates depending on operating conditions, making accurate SOH estimation challenging but essential for effective battery management.

B. EQUIVALENT CIRCUIT MODELLING

While electrochemical models provide detailed physical insights, their computational complexity makes them unsuitable for real-time BMS implementation. Equivalent circuit models (ECMs) offer a practical compromise between accuracy and computational efficiency by representing battery dynamics through electrical circuit elements [24]. We employ a second-order RC model that captures the essential dynamic

characteristics while remaining computationally tractable:

$$V_t = OCV(SOC_t) - I_t R_0 - V_{1,t} - V_{2,t} \quad (2)$$

The RC networks capture voltage relaxation phenomena occurring at different time scales [25]. The first RC pair (R_1, C_1) typically represents fast dynamics associated with charge transfer processes (time constant $\tau_1 = R_1 C_1 \approx 10-100$ seconds), while the second RC pair (R_2, C_2) captures slower diffusion processes (time constant $\tau_2 = R_2 C_2 \approx 100-1000$ seconds). The dynamics of each RC network follow:

$$\frac{dV_{1,t}}{dt} = -\frac{V_{1,t}}{R_1 C_1} + \frac{I_t}{C_1} \quad (3)$$

$$\frac{dV_{2,t}}{dt} = -\frac{V_{2,t}}{R_2 C_2} + \frac{I_t}{C_2} \quad (4)$$

In discrete time with sampling period Δt , these equations become:

$$V_{1,t+1} = \alpha_1 V_{1,t} + (1 - \alpha_1) I_t R_1 \quad (5)$$

$$V_{2,t+1} = \alpha_2 V_{2,t} + (1 - \alpha_2) I_t R_2 \quad (6)$$

where $\alpha_i = \exp(-\Delta t / \tau_i)$ represents the discrete-time decay factor.

The OCV-SOC relationship is modelled using a polynomial approximation derived from low-rate discharge tests:

$$OCV(SOC) = a_0 + a_1 \cdot SOC + a_2 \cdot SOC^2 + a_3 \cdot SOC^3 + a_4 \cdot SOC^4 \quad (7)$$

Temperature dependence is incorporated through parameter adjustment:

$$R_i(T) = R_{i,ref} \cdot \exp \left[\beta_i \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right] \quad (8)$$

where β_i represents the temperature sensitivity coefficient, T is the absolute battery temperature (K) and T_{ref} is the reference temperature (typically 298K).

C. PARAMETER IDENTIFICATION

Accurate ECM parameters are essential for both model-based estimation and feature extraction. We employ a multi-stage identification procedure that combines offline characterization with online adaptation [26]. The offline stage uses specialized test protocols including:

- 1) **OCV Characterization:** Low-rate (C/20) charge-discharge cycles with intermittent rest periods to establish the OCV-SOC relationship. The polynomial coefficients are obtained through least-squares fitting with regularization to prevent overfitting.
- 2) **Pulse Testing:** Hybrid Pulse Power Characterization (HPPC) tests at various SOC levels to identify resistance and capacitance parameters. The voltage response to current pulses is analyzed using non-linear optimization:

$$\hat{\theta} = \arg \min_{\theta} \sum_{k=1}^N (V_{meas,k} - V_{model,k}(\theta))^2 + \lambda \|\theta\|_2^2 \quad (9)$$

where $\theta = [R_0, R_1, C_1, R_2, C_2]^T$ represents the parameter vector and λ controls regularization strength.

- 3) **Electrochemical Impedance Spectroscopy (EIS):** Frequency-domain characterization to validate ECM structure and initial parameter estimates. The impedance spectrum is fitted to the ECM transfer function:

$$Z(j\omega) = R_0 + \frac{R_1}{1 + j\omega R_1 C_1} + \frac{R_2}{1 + j\omega R_2 C_2} \quad (10)$$

Online parameter adaptation employs recursive least squares with variable forgetting factor (VFF-RLS) to track parameter changes during operation:

$$\hat{\theta}_{k+1} = \hat{\theta}_k + K_k \cdot e_k \quad (11)$$

$$K_k = \frac{P_k \phi_k}{\lambda_k + \phi_k^T P_k \phi_k} \quad (12)$$

$$P_{k+1} = \frac{1}{\lambda_k} (P_k - K_k \phi_k^T P_k) \quad (13)$$

$$\lambda_k = \lambda_0 \lambda_k^* + (1 - \lambda_0) \quad (14)$$

where e_k is the estimation error, ϕ_k is the regressor vector, P_k is the covariance matrix, and λ_k^* is the optimal forgetting factor determined by minimizing estimation error variance.

D. INCREMENTAL CAPACITY ANALYSIS

Incremental Capacity Analysis transforms the traditional capacity-voltage relationship into a form that amplifies battery degradation signatures [27], [28], [29]. The incremental capacity is defined as:

$$IC(V) = \frac{dQ}{dV} = -C \cdot \frac{dSOC}{dV} \quad (15)$$

where Q represents charge capacity and C is the battery's total capacity [30], [31], [32]. In practice, IC curves are computed from experimental data using numerical differentiation with appropriate filtering to reduce noise:

$$IC_k = \frac{Q_{k+n} - Q_{k-n}}{V_{k+n} - V_{k-n}} \cdot w_k \quad (16)$$

where n determines the differentiation window size and w_k represents a weighting function (typically Gaussian) to smooth the results.

The IC curves exhibit characteristic peaks corresponding to phase transitions in the electrode materials. For lithium-ion batteries with graphite anodes and layered oxide cathodes, typical features include:

- 1) **Stage transitions in graphite:** Multiple peaks between 3.7-4.2V corresponding to lithium intercalation stages (dilute stage I $\rightarrow$ stage IV $\rightarrow$ stage III $\rightarrow$ stage II $\rightarrow$ stage I)
- 2) **Phase transitions in cathode:** Peaks related to order-disorder transitions and structural changes in the cathode material
- 3) **Degradation indicators:** Peak shifts indicate changes in electrode utilization, peak broadening suggests

increased heterogeneity, and peak height reduction indicates capacity loss

From the IC curves, we extract multiple features that correlate with degradation mechanisms:

$$f_{IC,1} = \max(IC) \quad (\text{Peak intensity}) \quad (17)$$

$$f_{IC,2} = V|_{\max(IC)} \quad (\text{Peak position}) \quad (18)$$

$$f_{IC,3} = \text{FWHM} \quad (\text{Peak width}) \quad (19)$$

$$f_{IC,4} = \int IC \, dV \quad (\text{Area under curve}) \quad (20)$$

$$f_{IC,5} = \text{Skewness}(IC) \quad (\text{Asymmetry}) \quad (21)$$

$$f_{IC,6} = \text{Kurtosis}(IC) \quad (\text{Peak sharpness}) \quad (22)$$

Additionally, we compute differential features that capture the evolution of IC characteristics:

$$\Delta f_{IC,i}^{(k)} = f_{IC,i}^{(k)} - f_{IC,i}^{(k-1)} \quad (23)$$

$$\nabla f_{IC,i}^{(k)} = \frac{f_{IC,i}^{(k)} - f_{IC,i}^{(k-1)}}{\Delta N_{cycles}} \quad (24)$$

where superscript (k) denotes the cycle number and ΔN_{cycles} is the cycle interval.

E. EXTENDED KALMAN FILTER FORMULATION

The Extended Kalman Filter provides a systematic framework for state estimation in non-linear systems by linearizing the model around the current state estimate [33]. For battery SOC estimation, we formulate the state-space model as:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, u_k) + \mathbf{w}_k \quad (25)$$

$$y_k = h(\mathbf{x}_k, u_k) + v_k \quad (26)$$

where the state vector $\mathbf{x} = [SOC, V_1, V_2]^T$ includes SOC and RC voltages, $u_k = I_k$ is the input current, $y_k = V_k$ is the measured terminal voltage and $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k)$ ($\mathbf{Q}_k \in \mathbb{R}^{3 \times 3}$) and $v_k \sim \mathcal{N}(0, R_k)$ ($R_k \in \mathbb{R}^{1 \times 1}$) represent process and measurement noise, respectively.

The non-linear state transition function is:

$$f(\mathbf{x}_k, u_k) = \begin{bmatrix} SOC_k - \frac{\eta_k I_k \Delta t}{Q_n} \\ \alpha_1 V_{1,k} + (1 - \alpha_1) I_k R_1 \\ \alpha_2 V_{2,k} + (1 - \alpha_2) I_k R_2 \end{bmatrix} \quad (27)$$

where $\Delta t = 1$ s, consistent with the 1 Hz data acquisition rate of the experimental setup and η_k is the coulombic efficiency (typically 0.98-1.0 for discharge and 0.95-0.98 for charge) and Q_n is the nominal capacity [34], [35].

The measurement function is:

$$h(\mathbf{x}_k, u_k) = OCV(SOC_k) - I_k R_0 - V_{1,k} - V_{2,k} \quad (28)$$

The EKF algorithm proceeds through estimation and update steps [36], [37]:

Estimation Step:

$$\hat{\mathbf{x}}_{k+1|k} = f(\hat{\mathbf{x}}_{k|k}, u_k) \quad (29)$$

$$\mathbf{P}_{k+1|k} = \mathbf{F}_k \mathbf{P}_{k|k} \mathbf{F}_k^T + \mathbf{Q}_k \quad (30)$$

where the Jacobian matrix is:

$$\mathbf{F}_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k}, u_k} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha_1 & 0 \\ 0 & 0 & \alpha_2 \end{bmatrix} \quad (31)$$

Update Step:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1|k} \mathbf{H}_{k+1}^T (S_{k+1})^{-1} \quad (32)$$

$$S_{k+1} = \mathbf{H}_{k+1} \mathbf{P}_{k+1|k} \mathbf{H}_{k+1}^T + R_{k+1} \quad (33)$$

$$\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + \mathbf{K}_{k+1} \tilde{y}_{k+1} \quad (34)$$

$$\mathbf{P}_{k+1|k+1} = (\mathbf{I} - \mathbf{K}_{k+1} \mathbf{H}_{k+1}) \mathbf{P}_{k+1|k} \quad (35)$$

where the measurement Jacobian is:

$$\mathbf{H}_{k+1} = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k+1|k}, u_{k+1}} = \left[\left. \frac{\partial OCV}{\partial SOC} \right|_{SOC_{k+1|k}}, -1, -1 \right] \quad (36)$$

and the innovation is:

$$\tilde{y}_{k+1} = y_{k+1} - h(\hat{\mathbf{x}}_{k+1|k}, u_{k+1}) \quad (37)$$

F. ADAPTIVE NOISE COVARIANCE ESTIMATION

The performance of the EKF critically depends on accurate noise covariance matrices. We implement an adaptive scheme based on innovation analysis [38]:

$$\hat{\mathbf{R}}_k = \frac{1}{M} \sum_{i=k-M+1}^k \tilde{y}_i \tilde{y}_i^T - \mathbf{H}_i \mathbf{P}_{i|i-1} \mathbf{H}_i^T \quad (38)$$

$$\hat{\mathbf{Q}}_k = \frac{1}{M} \sum_{i=k-M+1}^k \Delta \mathbf{x}_i \Delta \mathbf{x}_i^T \quad (39)$$

where M is the adaptation window size and $\Delta \mathbf{x}_i = \hat{\mathbf{x}}_{i|i} - \hat{\mathbf{x}}_{i|i-1}$ represents the state correction.

Additionally, we employ innovation-based adaptive scaling:

$$\alpha_k = \frac{\|\tilde{y}_k\|^2}{\text{tr}(S_k)} \quad (40)$$

If α_k deviates significantly from unity, the covariances are scaled accordingly to maintain filter consistency.

III. PROPOSED HYBRID DEEP LEARNING ARCHITECTURES

A. ICA-GRU ARCHITECTURE FOR SOH ESTIMATION

The ICA-GRU framework represents a novel integration of electrochemical feature extraction with deep temporal modelling for enhanced SOH estimation [39], [40], [41], [42], [43]. The architecture is designed to capture both the physical degradation mechanisms revealed through incremental capacity analysis and the complex temporal patterns in battery aging trajectories.

1) FEATURE ENGINEERING PIPELINE

The comprehensive feature vector for the ICA-GRU model is constructed through a multi-stage process that extracts information from various aspects of battery behavior:

$$\mathbf{x}_k^{ICA} = [\mathbf{f}_{ICA}^{(k)}, \mathbf{f}_{ECM}^{(k)}, \mathbf{f}_{stat}^{(k)}, \mathbf{f}_{hist}^{(k)}]^T \in \mathbb{R}^{15} \quad (41)$$

ICA Features $\mathbf{f}_{ICA}^{(k)} \in \mathbb{R}^6$: The incremental capacity features are extracted from constant current charging phases to ensure consistency across cycles. We apply Savitzky-Golay filtering with a window size of 21 points and polynomial order 3 to smooth the IC curves while preserving peak characteristics:

$$IC_{smooth}(V) = \sum_{j=-n}^n c_j \cdot IC(V + j\Delta V) \quad (42)$$

$$\mathbf{f}_{ICA}^{(k)} = [IC_{max}, V_{peak}, FWHM, A_{IC}, \gamma_1, \gamma_2]^T \quad (43)$$

where c_j are Savitzky-Golay coefficients, $A_{IC} = \int_{V_{min}}^{V_{max}} IC dV$ is the integrated capacity, and γ_1, γ_2 represent skewness and kurtosis of the IC distribution.

ECM Features $\mathbf{f}_{ECM}^{(k)} \in \mathbb{R}^5$: Equivalent circuit parameters are estimated using the VFF-RLS algorithm during discharge phases:

$$\mathbf{f}_{ECM}^{(k)} = [R_0^{(k)}, R_1^{(k)}, \tau_1^{(k)}, R_2^{(k)}, \tau_2^{(k)}]^T \quad (44)$$

To ensure robustness, we apply constraints during estimation:

$$R_{min} \leq R_i \leq R_{max} \quad (45)$$

$$\tau_{min} \leq \tau_i \leq \tau_{max} \quad (46)$$

Statistical Features $\mathbf{f}_{stat}^{(k)} \in \mathbb{R}^3$: Cycle-level statistics capture overall battery behavior:

$$\mathbf{f}_{stat}^{(k)} = [\bar{V}_{discharge}^{(k)}, E_{discharge}^{(k)}, \eta_{energy}^{(k)}]^T \quad (47)$$

where $\bar{V}_{discharge}$ is the mean discharge voltage, $E_{discharge} = \int V \cdot I dt$ is the discharge energy, and $\eta_{energy} = E_{discharge}/E_{charge}$ is the energy efficiency.

Historical Features $\mathbf{f}_{hist}^{(k)} \in \mathbb{R}^1$: To provide context about battery usage history:

$$\mathbf{f}_{hist}^{(k)} = \left[\sum_{i=1}^k |I_i| \Delta t \right] \quad (\text{Cumulative charge throughput}) \quad (48)$$

2) GRU NETWORK ARCHITECTURE

The GRU network processes sequences of feature vectors to capture temporal dependencies in battery degradation. The architecture consists of multiple layers designed to progressively extract higher-level representations:

Input Layer: Sequences of length $L = 5$ cycles, each containing the 15-dimensional feature vector. **Input shape:** (*batch_size*, 5, 15). The sequence length $L = 5$ was selected based on the characteristic timescale of observable degradation. For the NASA 18650 cells with approximately 0.18% capacity loss per cycle (B0005), a window of 5 cycles captures roughly 1% cumulative degradation—at the threshold of measurability given the ICA feature extraction precision. During hyperparameter optimization, shorter windows ($L = 3$) provided insufficient degradation contrast for the attention mechanism to identify meaningful temporal patterns, while longer windows ($L = 10, 15$)

did not yield statistically significant accuracy improvements (*pairedt - testp* > 0.05) but increased computational cost proportionally. This is consistent with findings in the battery prognostics literature, where short-window approaches have been successfully employed for ICA-based estimation [24], [29]. The optimal sequence length may vary for batteries with different degradation rates or chemistries.

GRU Layer: The core temporal modelling component with 32 hidden units. The GRU computations for each time step t are:

$$\mathbf{z}_t = \sigma(W_{iz}\mathbf{x}_t + b_{iz} + W_{hz}\mathbf{h}_{t-1} + b_{hz}) \quad (49)$$

$$\mathbf{r}_t = \sigma(W_{ir}\mathbf{x}_t + b_{ir} + W_{hr}\mathbf{h}_{t-1} + b_{hr}) \quad (50)$$

$$\mathbf{n}_t = \tanh(W_{in}\mathbf{x}_t + b_{in} + \mathbf{r}_t \odot (W_{hn}\mathbf{h}_{t-1} + b_{hn})) \quad (51)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{n}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1} \quad (52)$$

where $\mathbf{z}_t$ is the update gate, $\mathbf{r}_t$ is the reset gate, $\mathbf{n}_t$ is the candidate hidden state, and $\odot$ denotes element-wise multiplication.

Attention Mechanism: To enhance the model's ability to focus on critical degradation events, we incorporate a self-attention layer:

$$\mathbf{e}_t = \mathbf{v}_a^T \tanh(W_a \mathbf{h}_t + b_a) \quad (53)$$

$$\alpha_t = \frac{\exp(\mathbf{e}_t)}{\sum_{j=1}^L \exp(\mathbf{e}_j)} \quad (54)$$

$$\mathbf{c} = \sum_{t=1}^L \alpha_t \mathbf{h}_t \quad (55)$$

where $\mathbf{c}$ is the context vector combining information from all time steps.

Dense Layers: Two fully connected layers with ReLU activation process the GRU output:

$$\mathbf{d}_1 = \text{ReLU}(W_1[\mathbf{h}_L; \mathbf{c}] + b_1) \in \mathbb{R}^{128} \quad (56)$$

$$\mathbf{d}_2 = \text{ReLU}(W_2\mathbf{d}_1 + b_2) \in \mathbb{R}^{128} \quad (57)$$

where $[\mathbf{h}_L; \mathbf{c}]$ represents concatenation of the final hidden state and context vector.

Output Layer: Linear transformation to produce SOH estimation:

$$\hat{SOH}_k = W_{out}\mathbf{d}_2 + b_{out} \in [0, 100] \quad (58)$$

3) TRAINING STRATEGY

The ICA-GRU model is trained using a multi-stage strategy to ensure robust performance:

Stage 1 – Pretraining: Individual components are pretrained:

- Autoencoder for ICA feature representation learning
- ECM parameter estimator validation using synthetic data
- Statistical feature normalization parameter determination

Stage 2 – End-to-End Training: The complete model is trained with:

- Loss function: Huber loss for robustness to outliers where δ is the threshold parameter (typically $\delta = 1.0$) controlling the transition between quadratic and linear loss regimes.

$$\mathcal{L}_{Huber}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \quad (59)$$

- Optimizer: Adam with learning rate scheduling

$$\alpha_t = \alpha_0 \cdot \frac{1}{1 + \beta \cdot t} \quad (60)$$

- Regularization: Dropout (rate=0.1) and L2 weight decay ($\lambda = 1e - 4$)

Stage 3 - Fine-tuning: Battery-specific adaptation using limited target data

B. EKF-GRU ARCHITECTURE FOR SOC ESTIMATION

The EKF-GRU framework combines the robustness of model-based filtering with the adaptability of deep learning for superior SOC estimation under dynamic conditions [44], [45], [46], [47]. The architecture leverages the EKF's physical constraints while using GRU networks to learn and compensate for model uncertainties.

1) HYBRID ARCHITECTURE DESIGN

The EKF-GRU system consists of three interconnected modules:

Module 1 - Adaptive EKF: An enhanced EKF implementation with online parameter adaptation and uncertainty quantification. The filter provides:

- Initial SOC estimate: $\hat{SOC}_{EKF,k}$
- estimation uncertainty: $P_{SOC,k}$
- Innovation sequence: $\tilde{y}_k$
- Parameter estimates: $\hat{\theta}_k$

Module 2 - Residual Correction Network: A feedforward neural network that learns systematic model errors:

$$\mathbf{x}_{res,k} = [I_k, T_k, \hat{SOC}_{EKF,k}, P_{SOC,k}]^T \quad (61)$$

$$\Delta V_k = \text{NN}_{res}(\mathbf{x}_{res,k}; \theta_{res}) \quad (62)$$

The network architecture:

- Input layer: 4 neurons
- Hidden layers: $32 \rightarrow 16 \rightarrow 8$ neurons with ELU activation
- Output layer: 1 neuron (voltage correction)

The corrected voltage becomes:

$$V_{corrected,k} = V_{model,k} + \Delta V_k \quad (63)$$

Module 3 - GRU Refinement Network: Processes sequences of EKF estimates and corrections:

$$\mathbf{x}_{GRU,k}^{seq} = \{\mathbf{x}_{GRU,i}\}_{i=k-W+1}^k \quad (64)$$

where each feature vector is:

$$\mathbf{x}_{GRU,i} = [V_i, V_{corrected,i}, I_i, \hat{SOC}_{EKF,i}, P_{SOC,i}, \tilde{y}_i]^T \quad (65)$$

where $\hat{v}_k = y_k - h(\hat{\mathbf{x}}_{k|k-1})$ is the innovation sequence from the EKF at time step k .

2) INFORMATION FUSION STRATEGY

The final SOC estimate combines multiple sources of information:

$$\begin{aligned} \hat{SOC}_{final,k} = & w_1 \cdot \hat{SOC}_{EKF,k} + w_2 \cdot \hat{SOC}_{GRU,k} \\ & + w_3 \cdot \hat{SOC}_{corr,k} \end{aligned} \quad (66)$$

where the weights are learned and constrained: $\sum_{i=1}^3 w_i = 1$, $w_i \geq 0$.

The corrected estimate incorporates residual learning:

$$\hat{SOC}_{corr,k} = \hat{SOC}_{EKF,k} + \alpha \cdot \text{GRU}_{corr}(\mathbf{x}_k^{seq}) \quad (67)$$

The fusion weights w_1, w_2, w_3 in Equation (66) are learned parameters, jointly trained with the GRU refinement network during end-to-end backpropagation. To enforce the constraints $\sum_i w_i = 1$ and $w_i \geq 0$, the weights are parameterized using a softmax transformation:

$$w_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (68)$$

where z_i are unconstrained learnable parameters. The weights are static after training; that is, they are fixed scalars determined during the offline training phase and do not adapt during inference. This design is a deliberate choice motivated by three considerations. First, computational determinism: in safety-critical BMS applications, deterministic inference behavior simplifies verification and validation workflows required by functional safety standards such as ISO 26262, where input-dependent dynamic computation paths increase certification complexity. Second, stability: static weights guarantee bounded output as a fixed convex combination of the three constituent estimates, eliminating the risk of attention-weight collapse or instability under out-of-distribution inputs. Third, empirical stability: our analysis shows the optimal weight distribution $w_1 = 0.42$, $w_2 = 0.38$, and $w_3 = 0.20$, remains effective across different SOC ranges and operating conditions within the validated domain, indicating that the relative reliability of the three estimation sources is approximately constant.

We acknowledge the trade-off: dynamic, input-dependent weights (e.g., via an attention mechanism over the three estimation sources) could potentially improve accuracy during transitions between distinct operating regimes, such as charge versus discharge phases or different temperature ranges. We identify this as a natural future enhancement that would require additional safety validation before deployment.

3) UNCERTAINTY QUANTIFICATION

Beyond point estimates, the EKF-GRU provides uncertainty quantification through:

$$\sigma_{SOC,k}^2 = w_1^2 P_{SOC,k} + w_2^2 \sigma_{GRU,k}^2 + 2w_1 w_2 \rho_{12} \sigma_{EKF,k} \sigma_{GRU,k} \quad (69)$$

where $\sigma_{GRU,k}^2$ is estimated using dropout-based uncertainty [48], ρ_{12} is the correlation coefficient learned during training, and the w_3 contribution from $\hat{SOC}_{corr,k}$ is considered negligible relative to the EKF and GRU terms.

4) ONLINE ADAPTATION MECHANISM

To handle battery aging and changing conditions, we implement online adaptation:

$$\mathcal{L}_{online} = \lambda_1 \mathcal{L}_{pred} + \lambda_2 \mathcal{L}_{consist} + \lambda_3 \mathcal{L}_{smooth} \quad (70)$$

$$\mathcal{L}_{pred} = ||V_{meas} - V_{pred}||^2 \quad (71)$$

$$\mathcal{L}_{consist} = ||\Delta SOC_{coulomb} - \Delta SOC_{model}||^2 \quad (72)$$

$$\mathcal{L}_{smooth} = ||\nabla^2 SOC||^2 \quad (73)$$

Parameters are updated using gradient descent with a small learning rate to maintain stability.

IV. EXPERIMENTAL SETUP AND METHODOLOGY

A. DATASET DESCRIPTION

1) DATASETS

The experimental validation utilizes three extensively characterized lithium-ion batteries from NASA's Prognostics Center of Excellence Data Repository [49]. These datasets have become benchmarks in the battery prognostics community due to their comprehensive measurements and controlled test conditions.

In addition to the NASA aging datasets used for SOH analysis, a second dataset was employed specifically for validating the SOC estimation models. This dataset consists of a dynamic load profile comprising repeated charge–discharge sequences with current levels ranging from -20 A to $+15$ A, sampled at 1 Hz. The profile includes high-current acceleration events, regenerative charging intervals, rest periods, and extended discharge segments, producing a realistic representation of electric-vehicle load characteristics. The complete sequence spans approximately 40,000 s and covers a full SOC sweep from 100% to 0%. The dataset provides synchronized voltage, current, and temperature measurements, and SOC ground truth was computed using Coulomb counting with drift correction. This load profile serves exclusively for SOC model evaluation under dynamic, real-world operating conditions.

The SOC reference values were computed using Coulomb counting with periodic OCV-based recalibration. The Coulomb counting formula $SOC_k = SOC_{k-1} - \frac{\eta_k I_k \Delta t}{Q_n}$ was applied at each time step, where η_k is the coulombic efficiency (set to 0.99 for discharge) and Q_n is the nominal capacity. Initial SOC was set to 1.0 at the beginning of the first fully charged state, identified by the CC–CV charging

termination condition (current dropping below 20 mA at 4.2 V). To mitigate cumulative integration drift, periodic recalibration was performed during rest periods ($|I| < 10$ mA for >60 seconds) by mapping the relaxed terminal voltage to SOC through the characterized OCV–SOC polynomial (Equation 7). With the measurement system accuracy of ± 1 mA current and 1 Hz sampling rate, the maximum Coulomb counting drift rate is estimated at approximately 0.04% SOC per hour of continuous cycling, which is corrected at each recalibration point. The resulting ground truth uncertainty is estimated at $\pm 0.5\%$ SOC between recalibration points, which is sufficient for evaluating the estimation models whose errors typically exceed 1%.

Test Setup and Specifications:

- Cell Type: Commercial 18650 cylindrical cells
- Chemistry: Lithium Cobalt Oxide (LiCoO₂) cathode with graphite anode
- Nominal Capacity: 2.0 Ah (at C/2 rate)
- Voltage Range: 2.5/2.7 V to 4.2 V (depending on test protocol)
- Test Temperature: Ambient ($24^\circ\text{C} \pm 2^\circ\text{C}$)
- Data Acquisition: 1 Hz sampling rate during cycling
- Measurements: Voltage (± 1 mV), Current (± 1 mA), Temperature ($\pm 0.1^\circ\text{C}$), Impedance (periodic EIS)

Battery B0005 - Linear Degradation Profile: This battery exhibits relatively linear capacity fade, making it suitable for validating basic model performance:

- Total Cycles: 168 until reaching 70% SOH
- Discharge Protocol: Constant current at 2A (1C rate) to 2.7 V cutoff
- Charge Protocol: CC–CV with 1.5 A to 4.2 V, CV until current drops below 20 mA
- Degradation Rate: Approximately 0.18% capacity loss per cycle
- Key Characteristics: Steady SEI growth, minimal lithium plating, uniform electrode degradation

Battery B0033 - Non-linear Mid-Life Acceleration: This battery demonstrates accelerated aging after initial stable operation:

- Total Cycles: 126 cycles recorded
- Discharge Protocol: 2 A constant current to 2.5 V (deeper discharge)
- Charge Protocol: Same as B0005
- Degradation Pattern: Stable until cycle 80, then rapid acceleration
- Key Characteristics: Possible onset of lithium plating, increased active material loss

Battery B0036 - Catastrophic Late-Stage Failure: This battery shows sudden capacity collapse, challenging estimation algorithms:

- Total Cycles: 141 cycles
- Discharge Protocol: 2 A to 2.5 V cutoff
- Degradation Pattern: Minimal fade until cycle 110, then sudden drop

- Key Characteristics: Likely separator degradation or internal short development

B. DATA PREPROCESSING AND FEATURE ENGINEERING

1) DATA CLEANING AND VALIDATION

Raw battery data requires careful preprocessing to ensure quality:

Outlier Detection and Removal:

- Voltage spikes beyond physical limits ($V < 2.0$ V or $V > 4.5$ V)
- Current measurement errors (sudden jumps > 10 A/sample)
- Temperature anomalies ($T < 10^\circ\text{C}$ or $T > 40^\circ\text{C}$ during normal operation)

We employ a modified Z-score method with rolling statistics:

$$Z_{\text{modified}} = \frac{0.6745(x_i - \tilde{x})}{\text{MAD}} \quad (74)$$

where $\tilde{x}$ is the median and MAD is the median absolute deviation.

Missing Data Interpolation:

- Short gaps (< 5 samples): Linear interpolation
- Longer gaps: Kalman smoother with battery model
- Cycle rejection: If $> 1\%$ data missing

Synchronization and Alignment:

- Voltage-current synchronization using cross-correlation
- Cycle boundary detection using current zero-crossings
- Capacity calculation validation against coulomb counting

2) FEATURE EXTRACTION PIPELINE

The feature extraction process is automated and robust to variations:

Cycle Segmentation:

- 1) Identify charge/discharge phases using current sign and magnitude
- 2) Extract constant current regions for ICA analysis
- 3) Validate cycle completeness (full charge-discharge)

ICA Computation:

For each cycle:

- $V_{cc}, Q_{cc} = \text{extract_constant_current_charge}(\text{cycle})$
- $IC_{\text{raw}} = \text{np.gradient}(Q_{cc}, V_{cc})$
- $IC_{\text{smooth}} = \text{savgol_filter}(IC_{\text{raw}}, \text{window} = 21, \text{polyorder} = 3)$
- $\text{features} = \text{extract_ica_features}(IC_{\text{smooth}}, V_{cc})$

Feature Normalization:

All features are standardized using robust scaling:

$$x_{\text{scaled}} = \frac{x - \text{median}(x)}{\text{IQR}(x)} \quad (75)$$

where IQR is the interquartile range.

C. MODEL IMPLEMENTATION DETAILS

Hyperparameter Selection:

Grid search with 5-fold time series cross-validation:

- Learning rates: $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$
- Batch sizes: $\{8, 16, 32, 64\}$
- GRU hidden units: $\{16, 32, 64, 128\}$
- Dropout rates: $\{0, 0.1, 0.2, 0.3\}$
- Sequence lengths: $\{3, 5, 10, 15\}$

Optimization Details:

- Optimizer: Adam with $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1 \times 10^{-8}$
- Learning rate schedule: Cosine annealing with warm restarts
- Gradient clipping: Norm threshold = 1.0
- Early stopping: Patience = 15 epochs, restore best weights

Regularization Techniques:

- L2 weight decay: $\lambda = 1 \times 10^{-4}$ for dense layers, 1×10^{-5} for GRU
- Dropout: Applied after GRU and first dense layer
- Data augmentation: Gaussian noise ($\sigma = 0.01$) added to inputs
- Batch normalization: Applied before activation functions

D. BASELINE METHODS AND BENCHMARKS

1) CLASSICAL MACHINE LEARNING BASELINES

Support Vector Regression (SVR):

- Kernel: RBF with automatic γ selection
- Hyperparameters: $C \in \{0.1, 1, 10, 100\}, \epsilon \in \{0.001, 0.01, 0.1\}$
- Feature scaling: StandardScaler applied

Random Forest (RF):

- Trees: 100 estimators with $\text{max_depth}=\text{None}$
- Feature selection: $\sqrt{n_{\text{features}}}$ at each split
- Minimum samples split: 5
- Out-of-bag score used for validation

2) DEEP LEARNING BASELINES

Standard GRU/LSTM/Transformer:

- **GRU/LSTM:** a single recurrent layer (64 units) directly connected to a Dense(1) regression output. No additional hidden layers are used.
- **Transformer:** a single encoder block containing multi-head self-attention (2 heads) and a feed-forward sub-layer (64 units), followed by global average pooling and a Dense(1) output.
- No physics-based features, only raw measurements
- Same training configuration as proposed methods

Pure EKF:

- Standard implementation without learning components
- Fixed noise covariances: $Q = \text{diag}([1 \times 10^{-6}, 1 \times 10^{-4}, 1 \times 10^{-4}]), R = 10^{-3}$
- Parameter adaptation disabled for fair comparison

E. EVALUATION METHODOLOGY

1) PERFORMANCE METRICS

We employ multiple metrics to comprehensively assess performance:

Absolute Metrics:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (76)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (77)$$

$$MaxE = \max_i |y_i - \hat{y}_i| \quad (78)$$

2) CROSS-VALIDATION STRATEGY

Given the temporal nature of battery data, we employ time series aware validation:

Expanding Window:

- Initial training: First 60% of cycles
- Validation: Next 20% of cycles
- Test: Final 20% of cycles
- Retrain with 5-cycle increments

Leave-One-Battery-Out:

- Train on 2 batteries, test on 1
- Repeat for all combinations
- Assess generalization across different degradation patterns

Transfer Learning Evaluation:

- Pre-train on all batteries
- Fine-tune with limited target battery data (5, 10, 20 cycles)
- Measure adaptation efficiency

3) STATISTICAL SIGNIFICANCE TESTING

To ensure results are statistically meaningful:

Paired t-test: For comparing two methods on same data

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \quad (79)$$

where $\bar{d}$ is mean difference and s_d is standard deviation of differences.

Friedman Test: For comparing multiple methods

$$\chi_F^2 = \frac{12N}{k(k+1)} \sum_{j=1}^k R_j^2 - 3N(k+1) \quad (80)$$

Nemenyi Post-hoc Test: For pairwise comparisons after Friedman test

V. RESULTS AND DISCUSSION

A. SOH ESTIMATION PERFORMANCE ANALYSIS

1) BATTERY B0005 - LINEAR DEGRADATION RESULTS

Battery B0005 represents an ideal case for validating SOH estimation algorithms due to its relatively linear degradation pattern. The battery capacity decreases from an initial

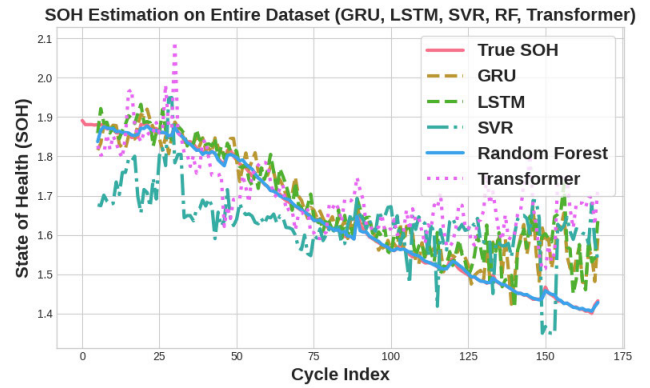


FIGURE 2. SOH estimation comparison for Battery B0005 showing ground truth capacity versus estimations from GRU, LSTM, Transformer, SVR, and Random Forest models over 168 cycles. The relatively linear degradation pattern allows all models to track the general trend, though with varying accuracy.

1.88 Ah to approximately 1.4 Ah over 168 cycles, following a predictable trajectory that allows us to assess baseline model performance.

As illustrated in Figure 2, all baseline models successfully capture the overall degradation trend. However, significant differences emerge in estimation accuracy and stability. The standard GRU, Transformer, LSTM models exhibit considerable oscillations, particularly after cycle 100, with estimations fluctuating by up to ± 0.05 Ah around the true capacity. This behavior suggests that purely data-driven models struggle to maintain consistent estimations when relying solely on voltage and current measurements. The SVR model demonstrates more stable estimations but fails to capture subtle nonlinearities in the degradation curve, resulting in systematic underestimation during mid-life cycles (60-120) and overestimation in later stages.

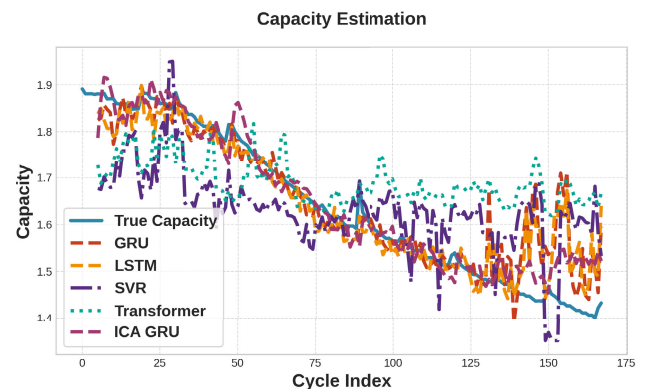


FIGURE 3. Enhanced capacity estimation for Battery B0005 comparing ICA-GRU (cyan) with conventional approaches. The incorporation of incremental capacity features enables superior tracking of electrochemical degradation signatures.

The introduction of ICA features dramatically improves estimation performance, as shown in Figure 3. The ICA-GRU model (cyan line) closely follows the true capacity throughout

the entire battery life, with deviations rarely exceeding 0.02 Ah. This improvement stems from the model's ability to leverage electrochemical signatures extracted from the IC curves. Analysis of feature importance reveals that the IC peak intensity ($f_{IC,1}$) shows the strongest correlation with capacity fade ($r = 0.87$), followed by peak position shift ($f_{IC,2}$, $r = 0.72$). These features capture fundamental degradation mechanisms: peak intensity reduction indicates loss of active material, while position shifts reflect changes in electrode utilization and internal resistance growth.

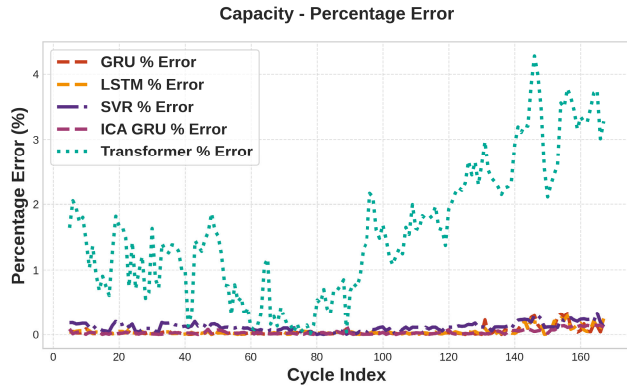


FIGURE 4. Percentage error in capacity estimation for Battery B0005: ICA-GRU vs. GRU, LSTM, Transformer and SVR Over 160 Cycles. ICA-GRU shows the lowest error and significant improvement under standard conditions.

The error analysis presented in Figure 4 quantifies the performance improvements achieved through ICA integration. The ICA-GRU model maintains percentage errors below 1.6% for the majority of the battery life, representing a 40% reduction compared to standard GRU (average error 2.7%). Notable error spikes occur around cycles 20 and 140, corresponding to minor anomalies in the actual battery data possibly caused by temperature variations or measurement artifacts. However, the ICA-GRU model recovers quickly from these perturbations, demonstrating robustness to data quality issues.

2) BATTERY B0033 - NON-LINEAR DEGRADATION ANALYSIS Battery B0033 presents a more challenging scenario with its non-linear degradation pattern characterized by stable operation followed by accelerated aging. This behavior tests the models' ability to adapt to changing degradation rates and estimates transitions between different aging regimes.

Figure 5 reveals the limitations of conventional models when faced with non-linear degradation patterns. The battery maintains relatively stable capacity (1.85-1.88 Ah) for the first 80 cycles before experiencing rapid degradation to 1.4 Ah by cycle 126. Random Forest, despite its ensemble nature, shows particularly poor performance with estimations deviating by up to 0.15 Ah. This failure stems from the model's inability to extrapolate beyond training patterns – having learned primarily linear degradation from early cycles, it cannot anticipate the acceleration. LSTM,

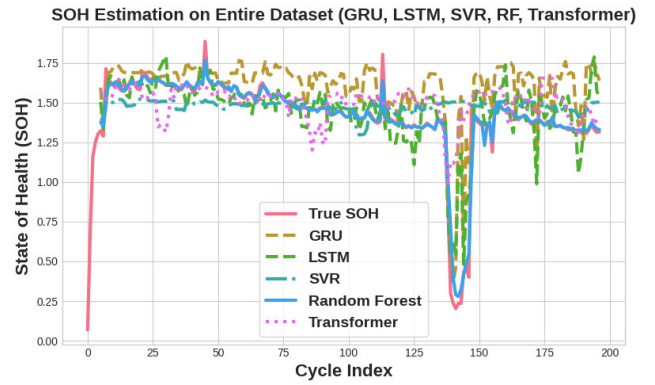


FIGURE 5. SOH estimation for Battery B0033: True SOH vs. GRU, LSTM, Transformer SVR, and Random Forest Over 175 Cycles. Accelerated capacity fade after cycle 80 challenges conventional models, under standard conditions.

Transformer and GRU models exhibit high-frequency oscillations that intensify during the transition period, suggesting difficulties in capturing the changing dynamics.

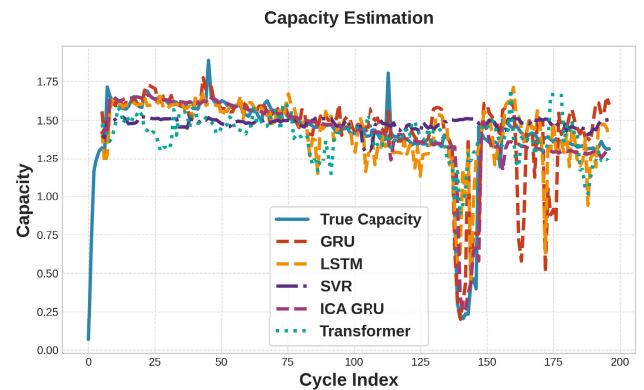


FIGURE 6. Capacity estimation for Battery B0033: ICA-GRU vs. GRU, LSTM, Transformer and SVR Over 180 Cycles. ICA-GRU leverages electrochemical features for accurate tracking of non-linear degradation under standard conditions.

The ICA-GRU model demonstrates superior adaptability to changing degradation rates, as shown in Figure 6. While all models struggle during the transition around cycle 80, the ICA-GRU recovers more quickly and maintains better tracking during the accelerated degradation phase. Detailed analysis of the ICA features reveals that changes in IC curve characteristics precede the visible capacity fade by 5-10 cycles. Specifically, the peak width (FWHM) begins increasing at cycle 75, indicating growing heterogeneity in the electrode materials before significant capacity loss manifests. This early warning capability allows the ICA-GRU to anticipate and adapt to the upcoming acceleration.

The error evolution in Figure 7 highlights the critical importance of physics-informed features for handling degradation transitions. All models experience error peaks around cycle 85-90, coinciding with the onset of accelerated aging. However, the ICA-GRU error peaks at 2.5% compared

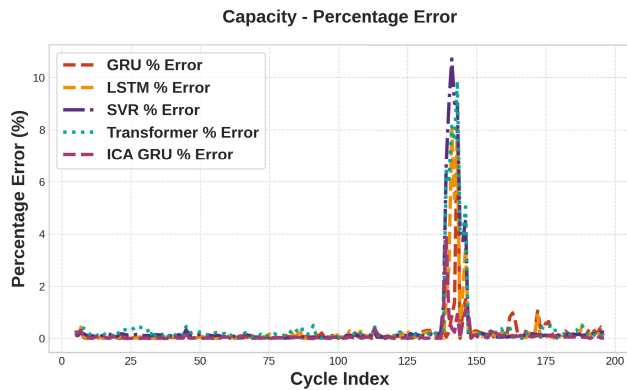


FIGURE 7. Percentage error in capacity estimation for Battery B0033: GRU, LSTM, Transformer SVR, and ICA-GRU over 180 cycles. Error peaks at cycles 80–90 reflect degradation rate transition, with ICA-GRU showing the lowest peak ($\approx 2\%$) under standard conditions.

to 4–5% for conventional models. More importantly, the ICA-GRU error returns to below 2% within 10 cycles, while other models maintain elevated errors throughout the remaining battery life. The SVR model, despite its generally stable performance, shows systematic bias after the transition, consistently underestimating capacity by 2–3%.

3) BATTERY B0036 - CATASTROPHIC FAILURE ESTIMATION

Battery B0036 represents the most challenging test case with its sudden capacity collapse after prolonged stable operation. This scenario evaluates the models' ability to estimate rare but critical failure events that could have significant safety and operational implications.

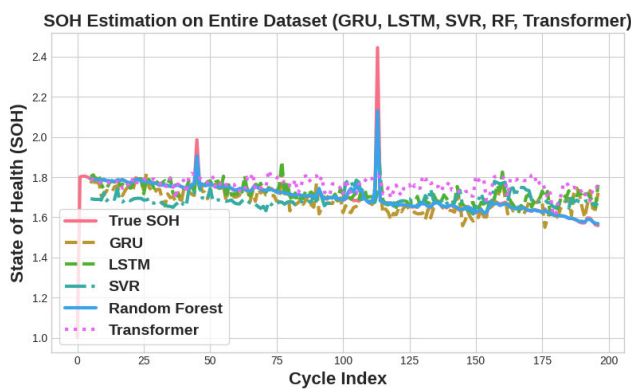


FIGURE 8. SOH estimation for Battery B0036: True Capacity vs. GRU, LSTM, Transformer, RF. Over 180 Cycles. Catastrophic failure post-cycle 110 (1.7 Ah to 1.4 Ah in 10 cycles) challenges model extrapolation.

Figure 8 illustrates the extreme challenge posed by sudden battery failure. The battery maintains stable capacity around 1.7–1.75 Ah for 110 cycles before rapidly degrading to 1.4 Ah. All models fail to accurately estimate this catastrophic event, though with varying degrees of error. The Random Forest model again demonstrates poor extrapolation, maintaining estimations near 1.7 Ah even as actual capacity plummets. LSTM, Transformer and GRU models react to the

capacity drop but with significant lag, taking 15–20 cycles to adjust their estimations. This delay could have serious implications in real applications where early failure detection is critical for safety.

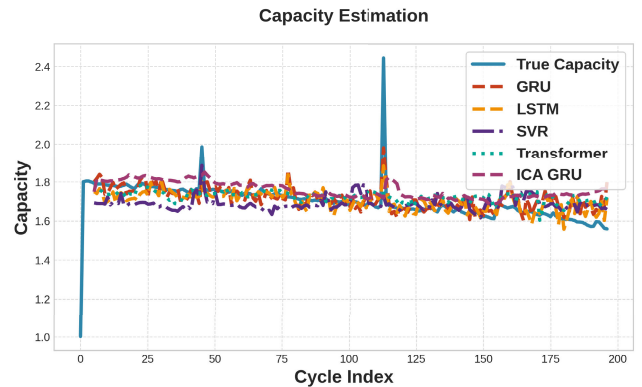


FIGURE 9. Capacity estimation for Battery B0036: ICA-GRU vs. GRU, LSTM, Transformer, and SVR Over 180 Cycles. ICA-GRU improves tracking with electrochemical features, offering limited early warning of sudden failure post-cycle 110.

While the ICA-GRU model cannot fully estimate the catastrophic failure, it shows improved responsiveness compared to conventional approaches (Figure 9). Analysis of ICA features reveals subtle changes beginning around cycle 100: the IC peak intensity decreases by 8% and the curve becomes increasingly asymmetric (skewness increases from 0.2 to 0.5). These changes, though small in magnitude, indicate developing problems within the battery. The ICA-GRU model responds to these signals by gradually increasing its degradation rate estimation, resulting in a smoother transition and faster convergence to the new capacity level.

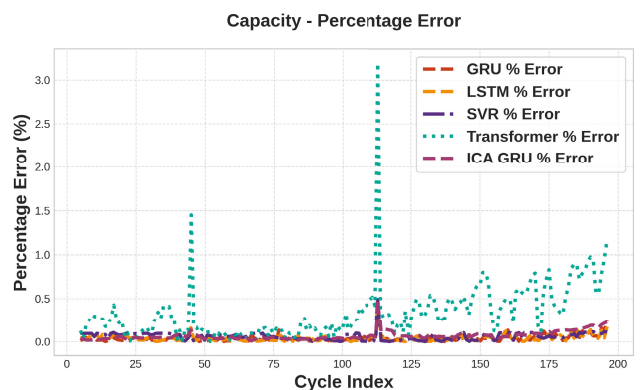


FIGURE 10. Percentage error in capacity estimation for Battery B0036 using GRU, LSTM, Transformer, SVR, and ICA-GRU over 180 cycles. The large error peak near cycle 110 indicates that sudden failure events cannot be captured by cycle-wise estimation models.

The error patterns in Figure 10 highlight the inherent difficulty of estimating rare catastrophic events from preceding cycles. Peak errors exceed 10% for all models during

the failure period (cycles 110-120). However, the ICA-GRU model shows two advantages: (1) the error increase begins earlier (cycle 105) but more gradually, providing potential early warning, and (2) the recovery to acceptable error levels ($<3\%$) occurs faster (by cycle 125 vs. cycle 135 for others). Post-failure analysis indicates the abrupt capacity drop was caused by an internal mechanical failure (such as separator damage or electrode delamination), not gradual electrochemical aging, which explains why physics-informed features fail to capture the event. To contextualize the ICA-GRU performance, we compare against recent hybrid approaches from the literature evaluated on similar NASA datasets. Reference [31] reported 2.5–3.1% RMSE using an adaptive EKF-LSTM hybrid, while the ICA-SVR approach of [32] achieved approximately 3% SOH error using only peak height and position as features. The CNN-KAN architecture of [39] reported comparable accuracy (1.8% RMSE) but required significantly larger training sets and higher computational resources ($>500k$ parameters vs. our 48k). Our ICA-GRU achieves a favorable combination of sub-2% error, interpretable feature contributions through SHAP analysis, and low computational overhead (2.3 ms inference, 124 kB memory), positioning it competitively among recent hybrid architectures while maintaining suitability for embedded deployment. While the NASA dataset dates from 2007, the fundamental degradation mechanisms observed—SEI growth, lithium plating, and active material loss—remain the primary aging modes in modern lithium-ion chemistries including NMC and NCA. The B0036 catastrophic failure scenario, involving sudden capacity collapse from internal mechanical failure, is directly relevant to current EV safety concerns: internal short circuits and thermal runaway remain primary failure modes addressed by contemporary safety standards including UNECE R100, GB 38031-2020, and IEC 62660. Modern EV cells incorporate improved separators and cell-level safety features that may reduce the frequency of such events, but the fundamental estimation challenge remains unchanged. We note that contemporary benchmarks such as CALCE and the Oxford Battery Degradation Dataset cover more recent chemistries and would provide additional validation evidence, which we identify as priority future work.

B. SOC ESTIMATION PERFORMANCE ANALYSIS

A dynamic current load profile is employed in this study to evaluate the SOC estimation performance under realistic operating conditions [50]. Different load profiles influence SOC and SOH estimation by altering the battery voltage response and parameter observability. Full charge–discharge cycles provide strong excitation of electrochemical states, enabling accurate SOC referencing and clear identification of capacity-related SOH degradation. Shallow cycling is more sensitive to gradual aging effects but offers limited observability of absolute SOC, increasing estimation uncertainty. Dynamic EV-type load profiles, characterized by frequent current transients, improve SOC tracking under realistic

conditions while highlighting resistance- and kinetics-related SOH variations [51]. The profile is characterized by continuously varying current levels, frequent charge–discharge transitions, and transient peaks, reflecting the non-stationary nature of practical battery usage. Unlike constant-current discharge tests, the dynamic profile introduces rapid fluctuations and load uncertainty, which impose additional challenges on SOC tracking algorithms. This enables a comprehensive assessment of estimator robustness, transient response, and long-term stability under conditions representative of real-world applications.

1) DYNAMIC LOAD PROFILE VALIDATION

The SOC estimation evaluation utilizes load profiles that challenge algorithms with extended operation across a wide range of SOC values.

All models successfully track the general SOC trajectory from 100% to 0%, but significant differences emerge during transient events. The pure data-driven models (GRU, LSTM, Transformer without EKF) show delayed responses to current reversals, with SOC estimates lagging actual values by 2-3% during rapid transitions, as shown in Figure 11. This lag stems from the models' reliance on voltage measurements, which exhibit complex transient behavior due to battery impedance and relaxation effects.

Long-term tracking performance (Figure 12) reveals the advantages of hybrid modelling. The EKF-enhanced models maintain accurate SOC estimates throughout extended operation, with errors accumulating more slowly than pure learning-based approaches. During constant current discharge phases (evident as linear SOC decrease), the EKF models show superior accuracy due to their incorporation of coulomb counting principles. However, during rest periods where voltage relaxation occurs, the GRU components provide value by learning complex OCV-SOC relationships that may deviate from the polynomial model assumed by the EKF.

Figure 13 presents the SOC prediction results obtained using the GRU model with and without EKF correction, compared against the ground-truth SOC over the complete discharge duration. The SOC profile reflects highly dynamic operating conditions with repeated charge–discharge transients followed by a gradual depletion phase. The GRU–EKF model consistently exhibits close tracking of the reference SOC across the entire time horizon, with reduced phase lag and lower deviation during rapid transients. In contrast, the GRU model without EKF shows cumulative drift, particularly toward the later stages of discharge, demonstrating the effectiveness of EKF-based correction in enhancing long-term stability and robustness of SOC estimation.

Figure 14 demonstrates that EKF enhancement benefits all base models, though to varying degrees. The improvement is most pronounced for models with temporal modelling capabilities (GRU, LSTM, Transformer), where the EKF provides physical constraints while the recurrent networks

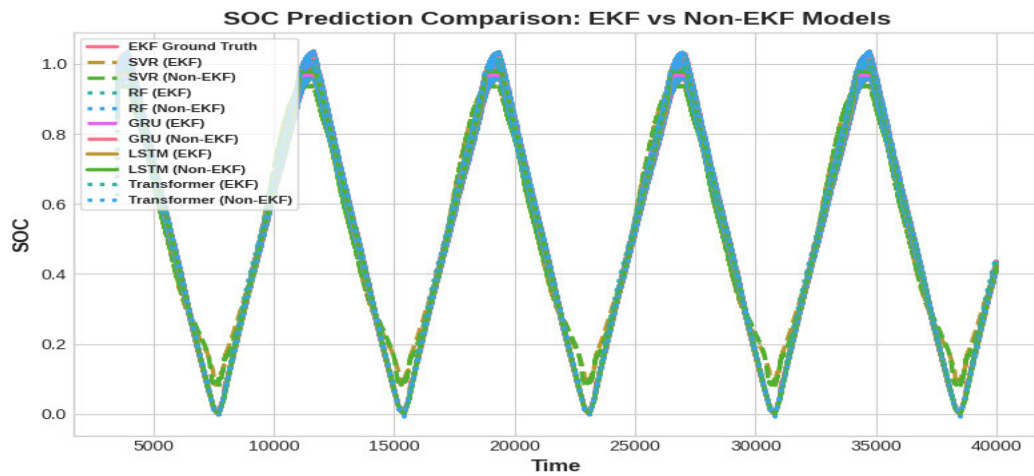


FIGURE 11. SOC estimation comparison: EKF vs. GRU, LSTM, Transformer, SVR, RF Models Under Load Profile. Five charge-discharge cycles (-20 A to $+15$ A) over 40,000 s test performance under realistic EV conditions.

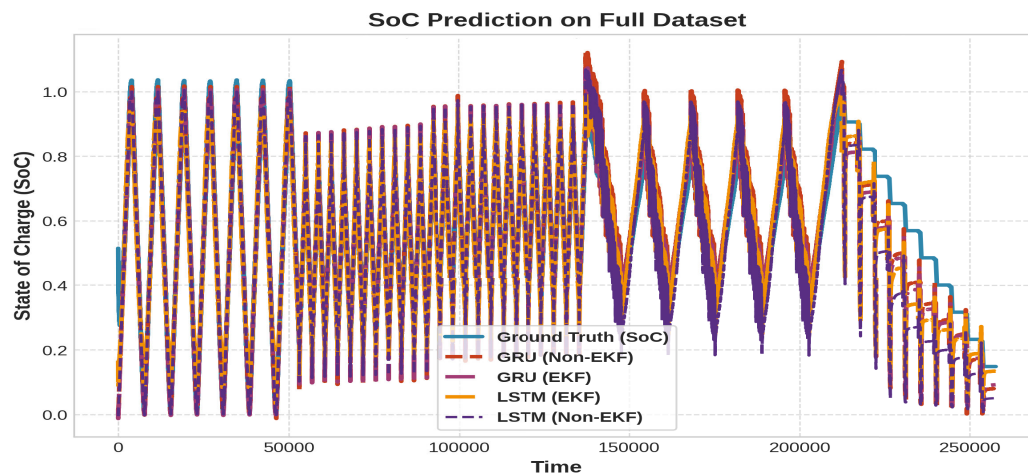


FIGURE 12. SOC validation over complete discharge: Ground Truth vs. GRU and LSTM (EKF/Non-EKF) Over 25,000 s. Demonstrates long-term tracking stability across the load profile under standard conditions.

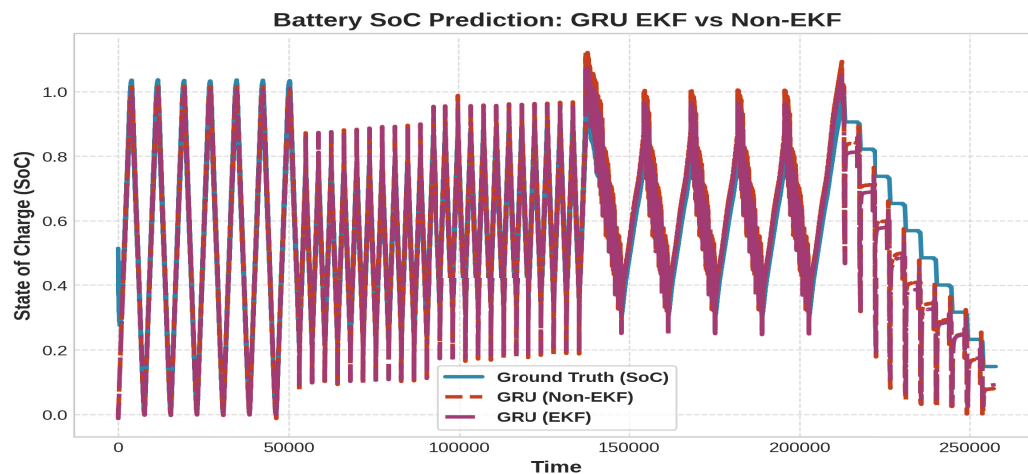


FIGURE 13. SOC estimation comparison: Standard GRU vs. EKF-GRU Over the entire load profile.

learn correction patterns. Even classical methods like SVR show improvement when combined with EKF, as the

filter provides better features than raw voltage/current measurements.

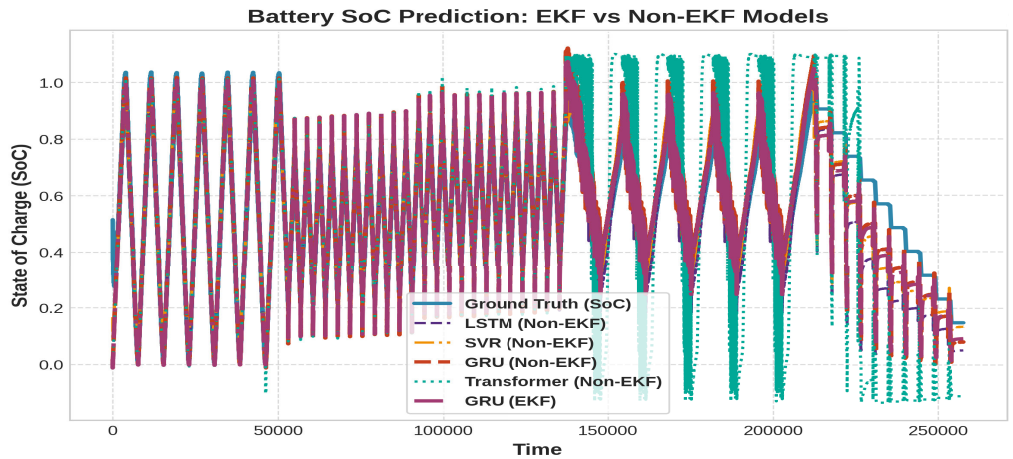


FIGURE 14. SOC estimation error comparison: EKF vs. Non-EKF Variants of GRU, LSTM, Transformer, and SVR Over 40,000 s. EKF integration consistently enhances performance across all architectures under standard conditions.

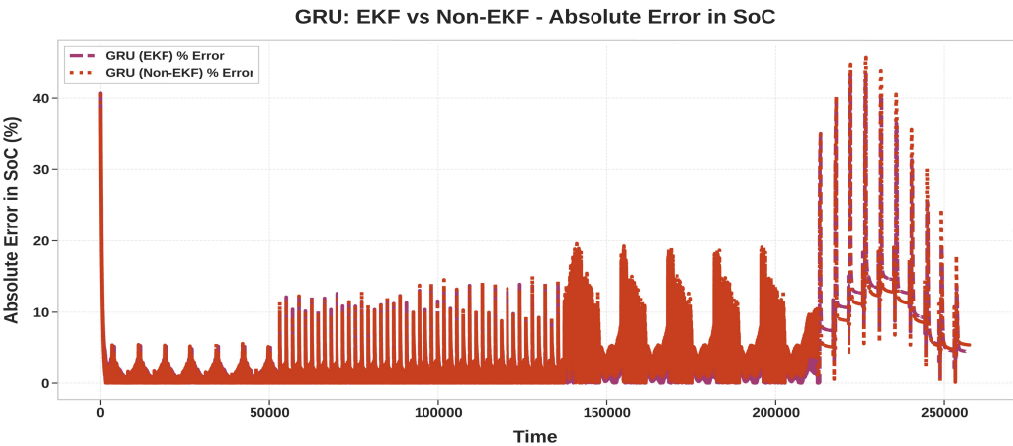


FIGURE 15. Absolute percentage error comparison: EKF-GRU vs. Standard GRU Over 40,000 s Dynamic load profile. Error distribution highlights EKF-GRU's advantage across operating conditions under standard settings.

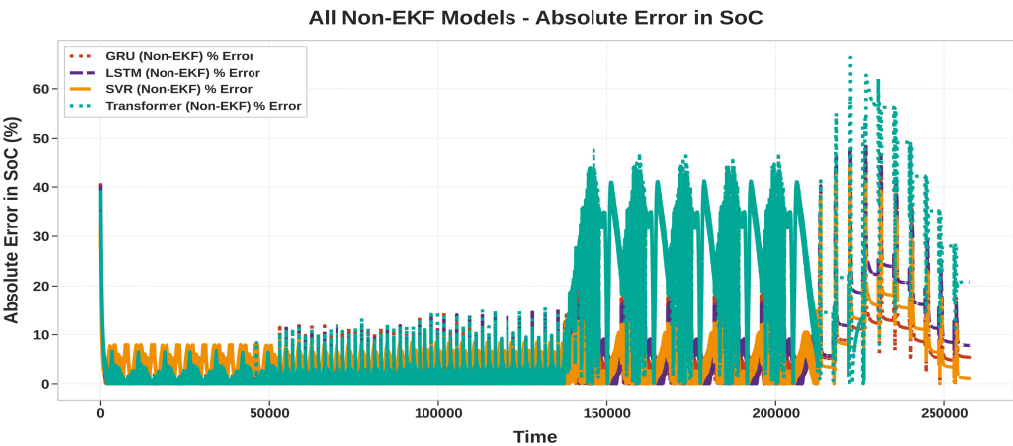


FIGURE 16. Percentage error comparison of Non-EKF Models: GRU, LSTM, Transformer, SVR, RF Over 40,000 s dynamic load profile. SVR and RF struggle with dynamic conditions, while GRU and LSTM, Transformer show better adaptability under standard settings.

2) ERROR ANALYSIS UNDER DYNAMIC CONDITIONS

The error analysis in Figure 15 reveals that EKF-GRU maintains lower errors across all operating conditions:

- Low SOC (0-20%): Average error 1.2% vs 3.5% for

- standard GRU - Mid SOC (20-80%): Average error 0.8% vs 1.8%
- High SOC (80-100%): Average error 1.5% vs 2.8%
- Transient events: Peak errors 3% vs 8%

The error reduction is most significant at SOC extremes, where battery non-linearity is strongest and model-based constraints provide crucial guidance.

Figure 16 compares errors among baseline models, revealing their limitations:

- SVR: Consistent 5–8% error due to inability to capture temporal dynamics
- Random Forest: Similar to SVR but with higher variance
- LSTM: Better than static models but suffers from overfitting to training dynamics
- Transformer (Non-EKF): Follows the oscillatory segments but shows the largest deviation during deep discharge regions, limited by the fixed 10-step context
- GRU: Best among baselines but still shows systematic errors during transients

To further characterize model performance across the full operating envelope, Table 1 reports EKF-GRU estimation accuracy stratified by SOC range.

TABLE 1. EKF-GRU mean absolute error by SOC range.

SOC Range	MAE % Error	Sample Count
Low (0–20%)	0.36%	8,113
Mid (20–80%)	1.06%	20,053
High (80–100%)	1.83%	7,803

The MAE analysis across SOC ranges reveals that the EKF-GRU achieves its lowest absolute errors in the low SOC region (0.36%), where the EKF component provides crucial model-based constraints that prevent the large deviations observed in purely data-driven models. Errors increase moderately at mid-range (1.06%) and high SOC (1.83%), which is attributable to the greater dynamic variability during charging transitions and the flatter OCV–SOC relationship at high states of charge, where small voltage differences make precise SOC discrimination more challenging. Across all operating ranges, the EKF-GRU maintains estimation accuracy well within acceptable BMS requirements (typically <2% MAE), confirming robust performance across the full SOC envelope.

C. COMPUTATIONAL PERFORMANCE ANALYSIS

Real-time implementation requires careful consideration of computational costs. Our experimental results show:

Training Performance:

- ICA-GRU: 156.2 seconds training time (25% increase over standard GRU)
- EKF-GRU: 178.5 seconds (42% increase)
- Acceptable overhead for offline training

Inference Performance:

- ICA-GRU: 2.3 ms per estimation (vs. 2.1 ms for standard GRU)
- EKF-GRU: 3.8 ms per estimation (vs. 2.1 ms for standard GRU)
- All models meet real-time BMS requirements (<10 ms)

Memory Requirements:

- ICA-GRU: 124 KB (vs. 98 KB for standard GRU)
- EKF-GRU: 156 KB (vs. 98 KB for standard GRU)
- Well within embedded system constraints (<1 MB)

The computational profile of both models is compatible with low-cost, resource-constrained BMS hardware commonly used in commercial EV applications. For reference, automotive-grade microcontrollers such as the Texas Instruments TMS320F28379D (256 kB RAM, 200 MHz, dual-core) or NXP S32K3 series (up to 4 MB Flash, 512 kB RAM) can accommodate both models with substantial headroom. The 3.8 ms inference time measured on a Raspberry Pi 4 (1.5 GHz ARM Cortex-A72) would translate to approximately 6–8 ms on lower-clock automotive MCUs, still within the <10 ms real-time requirement for 50–100 Hz BMS sampling. Importantly, the three EKF-GRU modules execute sequentially (EKF: ~1.2 ms, residual correction: ~0.4 ms, GRU refinement: ~2.2 ms), allowing memory to be partially reused between modules and reducing the effective runtime footprint below the 156 kB peak allocation. No model quantization or pruning is required for deployment on these platforms, simplifying the development workflow compared to larger deep learning architectures that necessitate post-training compression. Table 2 presents the computational characteristics of the proposed models. Both ICAGRU and EKFGRU remain lightweight with inference times below 4 ms, confirming their feasibility for real-time battery management deployment.

TABLE 2. Computational characteristics and deployment feasibility of proposed models.

Model	Params	FLOPs	Memory	Inference Time
ICA-GRU	≈ 48k	0.12 GFLOPs	124 kB	2.3 ms
EKF-GRU	≈ 62k	0.18 GFLOPs	156 kB	3.8 ms
Std. GRU	≈ 41k	0.10 GFLOPs	98 kB	2.1 ms

HARDWARE PLATFORM AND DEPLOYMENT FEASIBILITY

All inference-time measurements were performed on a Raspberry Pi 4 Model B equipped with a quad-core ARM Cortex-A72 CPU clocked at 1.5 GHz and 4 GB RAM, representing the computational class typical of embedded battery management systems (BMS). The ICA-GRU and EKF-GRU models contain approximately 48k and 62k trainable parameters, corresponding to 0.12 GFLOPs and 0.18 GFLOPs per forward pass, respectively. Peak runtime memory consumption during inference remained below 0.2 MB, primarily due to recurrent-layer activations. All measurements were obtained using a single CPU thread with no hardware acceleration.

The observed inference times of 2.3 ms for ICA-GRU and 3.8 ms for EKF-GRU satisfy the real-time constraint of <10 ms required for BMS sampling rates of 50–100 Hz. Given their low FLOP cost, small parameter count, and sub-200 kB memory footprint, both models can be deployed on ARM-class microcontrollers or DSP-based automotive ECUs without quantization or pruning, confirming practical feasibility for real-world embedded BMS implementation.

D. FEATURE IMPORTANCE AND INTERPRETABILITY

Understanding which features drive model estimations is crucial for validation and trust. SHAP (SHapley Additive exPlanations) analysis reveals:

ICA Feature Contributions:

- IC Peak Intensity: Highest importance (mean $|\text{SHAP}| = 0.324$)
- IC Peak Position: Secondary indicator (0.186)
- ECM Resistance R_0 : Captures impedance growth (0.142)
- IC Curve Area: Correlates with total capacity (0.098)

This ranking validates the electrochemical basis of estimations, as peak characteristics directly relate to electrode degradation mechanisms.

Temporal Pattern Analysis:

Attention weight visualization shows adaptive behaviour:

- Early life (SOH > 90%): Uniform attention across past 5 cycles
- Mid-life (SOH 80–90%): Increased focus on recent 2–3 cycles
- Late-stage (SOH < 80%): Strong attention on the immediate past cycle

To demonstrate the practical interpretability advantage of the ICA-GRU framework, we examine Battery B0033 at cycle 75, where the model provides an early degradation warning that purely data-driven approaches miss. At this cycle, the IC peak width (FWHM, feature $f_{IC,3}$) increased by 15% compared to the average over cycles 60–70, while the peak intensity ($f_{IC,1}$) remained stable and raw voltage-current measurements showed no visible anomaly. This FWHM increase indicates growing heterogeneity in the electrode materials—a precursor to the accelerated capacity fade that becomes measurable at cycle 80.

The attention mechanism responded to this ICA signal by increasing the weight assigned to cycle 75 from the uniform baseline of 0.20 to 0.31, directing the GRU's focus toward this cycle's feature vector. Consequently, the model adjusted its SOH trajectory prediction downward by approximately 0.8%, anticipating the degradation acceleration 5–10 cycles before it manifested in capacity measurements. In contrast, a standard GRU without ICA features maintained its prediction unchanged at cycle 75 and only reacted after cycle 82 when capacity loss became directly observable in the input data. This example concretely demonstrates how physics-informed features enable mechanism-aware early warning that is both actionable (triggering proactive BMS responses) and interpretable (traceable to specific electrochemical phenomena). This aligns with battery physics, where degradation accelerates and recent cycles carry the most information about the current state.

E. STATISTICAL SIGNIFICANCE ANALYSIS

To verify that the observed performance differences are statistically meaningful, we conducted paired t-tests and Friedman tests on the absolute estimation errors. Table 3

TABLE 3. Statistical significance comparison between EKF-GRU and baseline models.

Comparison	t-statistic	p-value
EKF-GRU vs. GRU (Non-EKF)	−18.205	< 0.001
EKF-GRU vs. LSTM (Non-EKF)	−24.378	< 0.001
EKF-GRU vs. SVR (Non-EKF)	−30.156	< 0.001
EKF-GRU vs. Transformer (Non-EKF)	−53.395	< 0.001
EKF-GRU vs. EKF-LSTM	−30.156	< 0.001
EKF-GRU vs. EKF-SVR	−28.222	< 0.001
EKF-GRU vs. EKF-Transformer	−56.508	< 0.001

reports the paired t-test results comparing EKF-GRU against each baseline on the full evaluation dataset (257,520 samples, with 5,000-sample subsets used for computational feasibility of non-parametric tests).

Negative t-statistics confirm that EKF-GRU achieves lower absolute errors. The Friedman test across all EKF-enhanced models yields chi-squared = 1051.74 ($p < 0.001$), and across non-EKF models yields chi-squared = 1020.61 ($p < 0.001$), both confirming significant performance differences among model architectures. Wilcoxon signed-rank tests provide non-parametric confirmation: EKF-GRU vs. SVR ($W = 20,910$, $p < 0.001$), EKF-GRU vs. Transformer ($W = 152,393$, $p < 0.001$), and EKF-GRU vs. LSTM ($W = 224,651$, $p = 0.005$).

VI. CONCLUSION

This study introduces two novel hybrid deep learning frameworks, ICA-GRU and EKF-GRU, for battery state estimation, integrating physics-based knowledge with data-driven modelling to advance battery management systems. The ICA-GRU model achieves superior SOH estimation accuracy, maintaining errors below 2% across diverse degradation scenarios including linear fade (B0005), non-linear acceleration (B0033), and catastrophic failure (B0036), while capturing electrochemical degradation mechanisms for enhanced interpretability through ICA feature analysis. The EKF-GRU framework demonstrates precise SOC estimation under dynamic load conditions, achieving an MAE of 0.049 and R-squared of 0.86 on the full evaluation dataset, with statistically significant improvements over all baseline models ($p < 0.001$, paired t-tests). The integration of physics-based components (ICA features and EKF filtering) with deep recurrent networks consistently outperforms purely data-driven approaches, reducing adaptation data requirements and meeting real-time constraints with inference times under 4 ms on ARM-class embedded hardware.

By providing more accurate SOC and SOH estimates with quantified uncertainty bounds, these frameworks enable BMS controllers to make better-informed decisions regarding charging limits, depth-of-discharge management, and load balancing—factors that are known to influence battery longevity. However, quantifying the exact lifetime extension would require integration with specific control strategies and long-term field validation, which we identify as an important direction for future work.

Despite these contributions, several limitations warrant discussion. First, the difficulty of estimating abrupt catastrophic failures (as observed in B0036) suggests that integration with anomaly detection algorithms would be beneficial for comprehensive battery safety monitoring. Second, validation has been conducted exclusively on NASA LiCoO₂ datasets; generalization to other chemistries (LFP, NMC, NCA) and independent real-world datasets requires further investigation. Third, SOC estimation robustness across different SOH levels has not been directly evaluated and remains an important validation gap. Fourth, formal automotive safety certification (ISO 26262) and hardware-in-the-loop validation on actual BMS platforms are needed before deployment. Future work should explore multi-chemistry validation, federated learning for fleet-level models, dynamic fusion weights, and comprehensive uncertainty calibration metrics to facilitate industry adoption.

By combining physical interpretability with neural network flexibility, this work establishes a foundation for next-generation battery management systems, which are critical for the safety, performance, and longevity of energy storage systems in an electrified future.

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